Plasma physics – the fourth state of matter

More than 99 % of the visible matter in the universe is plasma: a gas in which the atoms have lost electrons, so that free ions and electrons move among one another. Stars, the aurora, lightning, neon tubes and fusion reactors are all plasmas. Because the particles are charged, the plasma responds as a whole to electric and magnetic fields, and that gives physics quite different from that of ordinary gases. This page covers the basic ideas of plasma physics – with two interactive figures and fusion research as the main example.

What is a plasma?

In 1879 William Crookes called gas discharges “radiant matter” – a fourth state. The name *plasma* came in 1928 from Irving Langmuir, who studied electrical discharges in gases at General Electric; he felt that the glowing gas carried electrons and ions around the way blood plasma carries blood cells.

A gas becomes a plasma when a significant fraction of its atoms is ionised. In thermal equilibrium the degree of ionisation is given by the Saha equation: although the ionisation energy of hydrogen is 13.6 eV, hydrogen is almost completely ionised already at about 10,000 K, because there are so many more free states available to a detached electron. But ionisation alone does not make an ionised gas a plasma. Three conditions are required:

1. Small Debye length

The plasma must be much larger than the Debye length λ_D, so that on the whole it is electrically neutral (quasi-neutral).

2. Many particles in the Debye sphere

The plasma parameter $N_D = \frac{4}{3}\pi n \lambda_D^3$ must be much larger than 1, so that collective behaviour dominates over individual collisions.

3. Few collisions

The plasma frequency must exceed the collision frequency, so that the particles have time to respond collectively before they hit neutral atoms.

Debye shielding

Put a positive charge into a plasma, and the electrons gather around it while the ions are pushed slightly away. The charge is shielded. Thermal motion prevents the shielding from being complete, and the result is that the potential does not fall off as $1/r$ but exponentially:

$$\phi(r) = \frac{q}{4\pi\varepsilon_0 r}\, e^{-r/\lambda_D}, \qquad \lambda_D = \sqrt{\frac{\varepsilon_0 k_B T_e}{n_e e^2}}$$

Peter Debye and Erich Hückel derived the formula in 1923 for electrolytes. Beyond a few Debye lengths the charge is not felt at all. In a tokamak λ_D is about 0.1 mm, in the solar wind about 10 m, and in the interstellar medium several metres.

The plasma frequency

If the electrons are displaced slightly relative to the ions, the electric field pulls them back, they overshoot and oscillate. The oscillation takes place at the *plasma frequency*, which depends only on the density:

$$\omega_p = \sqrt{\frac{n_e e^2}{\varepsilon_0 m_e}}, \qquad f_p \approx 9\ \mathrm{Hz} \cdot \sqrt{n_e / \mathrm{m}^{-3}}$$

Electromagnetic waves with a frequency below f_p cannot penetrate the plasma; they are reflected. That is why short-wave radio can be sent round the Earth by reflection in the ionosphere, where f_p is a few MHz, while FM radio and GPS signals pass straight through. Metals are shiny for the same reason: the electrons in a metal are a plasma with f_p in the ultraviolet.

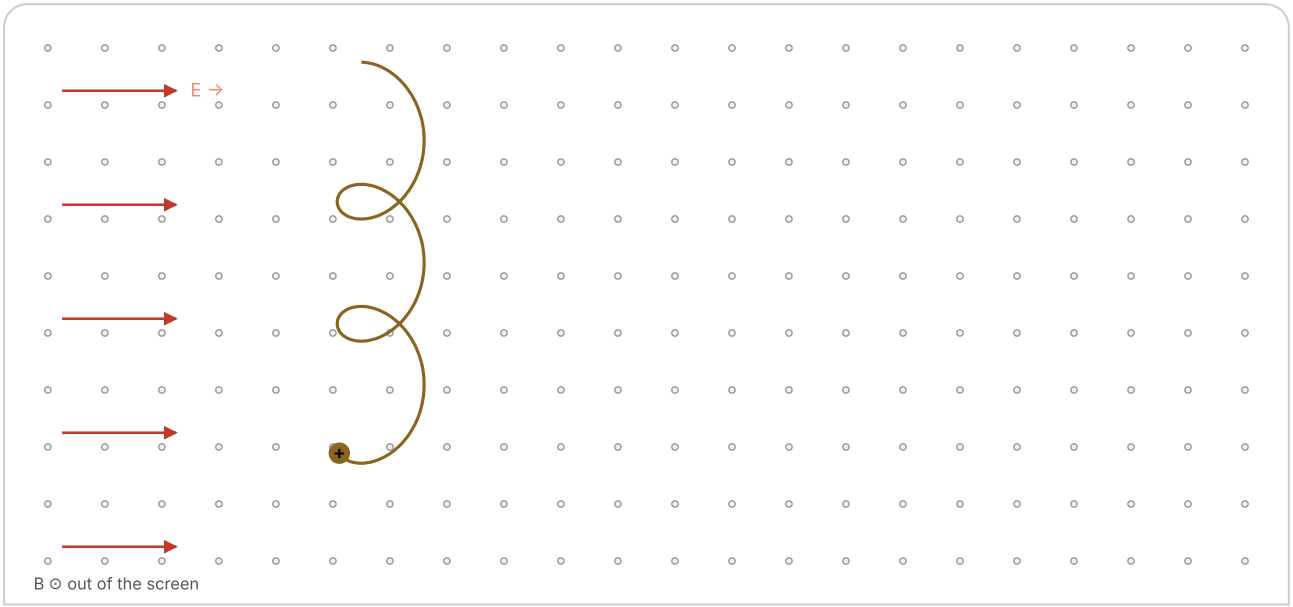
Charged particles in fields

In a magnetic field a charged particle moves in a helix around the field lines with the cyclotron frequency and Larmor radius

$$\omega_c = \frac{|q|B}{m}, \qquad r_L = \frac{m v_\perp}{|q|B}$$

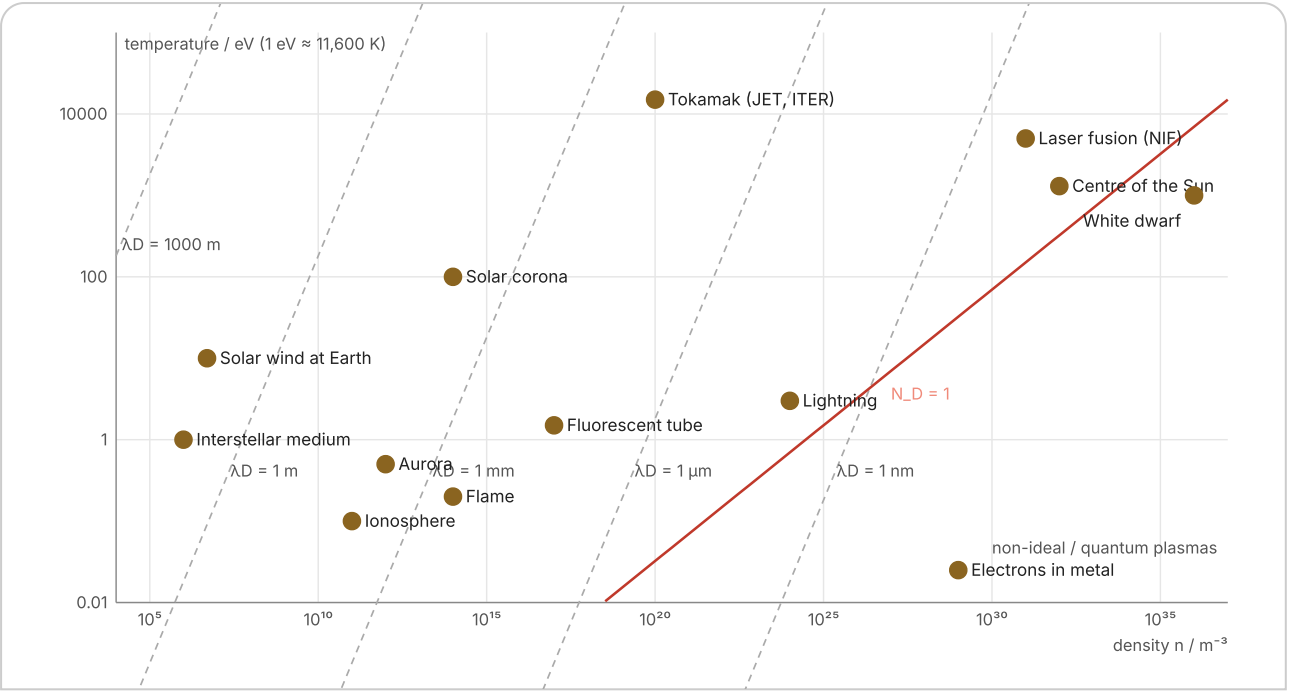
Electrons gyrate quickly in small circles, ions slowly in large ones. If there is also an electric field perpendicular to $\vec{B}$, the centre of the circle drifts with a velocity that is the same for all particles regardless of charge and mass – the *E × B drift*. A magnetic field that varies in strength gives a drift that depends on the charge and therefore separates ions and electrons – and drives currents. These drifts are the key to understanding the aurora, the Van Allen belts and the problems of confining a fusion plasma.

$$\vec{v}_E = \frac{\vec{E} \times \vec{B}}{B^2}, \qquad \vec{v}_{\nabla B} = \frac{m v_\perp^2}{2qB^3}\, \vec{B} \times \nabla B$$



Plasmas in nature and in the laboratory

Plasmas span more than 30 orders of magnitude in density and 7 in temperature. Even so, they can be described with the same concepts.



Plasma as a fluid: magnetohydrodynamics

On large scales a plasma can be described as an electrically conducting fluid. In 1942 Hannes Alfvén realised that in a well-conducting plasma the magnetic field lines are “frozen into” the matter: they move with the flow, and the flow follows the field lines. He predicted a new type of wave in which the field lines vibrate like the strings of a guitar, with the Alfvén speed

$$v_A = \frac{B}{\sqrt{\mu_0 \rho}}$$

Alfvén waves were demonstrated in liquid mercury in 1949 and have since been measured in the solar wind and the Sun’s corona. Alfvén received the Nobel Prize in 1970 for magnetohydrodynamics. The frozen-in field explains why the Sun’s magnetic field is drawn out by the solar wind into a spiral throughout the solar system, and why sunspots appear in pairs.

Waves and instabilities

A plasma can carry far more types of wave than a neutral gas: electron plasma waves, ion-acoustic waves, Alfvén waves, cyclotron waves and many more. In 1946 Lev Landau showed that plasma waves can be damped even without collisions, because particles moving at almost the wave speed exchange energy with it. Landau damping sounded so paradoxical that it was only accepted once it had been measured in 1964.

Plasmas are also notorious for instabilities. A plasma held by magnetic fields is like a liquid balanced on an elastic membrane: it finds every weakness. Kink and sausage instabilities,

Rayleigh–Taylor instabilities and microscopic turbulence have been the central problem of fusion research for 70 years.

Fusion: the Sun’s energy on Earth

The easiest fusion reaction to start is between the two heavy hydrogen isotopes deuterium and tritium:

$$\mathrm{D} + \mathrm{T} \;\rightarrow\; {}^{4}\mathrm{He}\,(3.5\ \mathrm{MeV}) + n\,(14.1\ \mathrm{MeV})$$

The reaction requires temperatures of about 150 million degrees – ten times hotter than the centre of the Sun, because we do not have the Sun’s enormous density and time. In 1955 John Lawson showed that a fusion plasma can only sustain itself (“ignite”) if the product of density, temperature and confinement time is large enough:

$$n T \tau_E \gtrsim 3 \cdot 10^{21}\ \mathrm{keV\,s\,m^{-3}}$$

There are two main routes. In *magnetic confinement* a thin plasma ($n \sim 10^{20}$ m⁻³) is held for several seconds by magnetic fields. In *inertial confinement* a small capsule is compressed by powerful lasers to hundreds of times solid density for less than a billionth of a second. In December 2022 the National Ignition Facility in California for the first time got more fusion energy out (3.15 MJ) than laser energy in (2.05 MJ). See also Nuclear physics.

The tokamak and JET

A plasma in a straight magnetic field escapes at the ends. Bend the field into a ring, and the particles drift out because of the gradient and curvature drifts. The solution is to twist the field lines so that each one runs both the long way and the short way round the ring, and the drifts average out. In the *tokamak*, developed in the Soviet Union by Igor Tamm and Andrei Sakharov in the 1950s and made famous by the results from T-3 in 1968, the twist comes from a large current flowing in the plasma itself. In the *stellarator*, invented by Lyman Spitzer in 1951, the field is twisted by complicated external coils; Wendelstein 7-X in Greifswald is the largest today.

The European JET (Joint European Torus) at Culham near Oxford had its first plasma in 1983 and was for 40 years the world’s largest tokamak. Scientists from all over Europe contributed, including the plasma physics department at Risø in Denmark. JET was the first machine to run on deuterium–tritium fuel, in 1991. In 1997 it reached 16 MW of fusion power, and in its very last experiments in 2023 it set a world record of 69 MJ of fusion energy from a single plasma discharge, before being shut down in December 2023. JET’s data are the basis for ITER, being

built at Cadarache in southern France, which is to produce ten times more fusion power than is supplied to the plasma.

	Centre of the Sun	JET / ITER	Laser fusion (NIF)
Temperature	15 million K	100–200 million K	50–100 million K
Density	10^{32} m^{-3}	10^{20} m^{-3}	10^{31} m^{-3}
Confinement time	billions of years	seconds	under 1 ns
Confinement by	gravity	magnetic field	inertia of matter

Plasma in space

The Sun's corona is over a million degrees – much hotter than the surface – and why is still not fully understood; Alfvén waves and small magnetic "reconnections" are the best candidates. From the corona the solar wind streams out at 400–800 km/s. It compresses the Earth's magnetic field on the day side and stretches it into a long tail on the night side. When the field in the tail reconnects, electrons are accelerated down along the field lines and make the oxygen and nitrogen atoms of the atmosphere glow: the aurora. Kristian Birkeland predicted around 1900 that the aurora is caused by currents along the field lines – an idea that was only confirmed by satellite measurements in the 1960s. See Aurora and The Sun.

Plasma technology is everywhere in daily life too: fluorescent tubes and plasma screens, welding and cutting, etching of microchips, sterilisation and the ion thrusters of spacecraft.

Further reading

- Francis F. Chen: *Introduction to Plasma Physics and Controlled Fusion* (3rd ed., Springer 2016)
- Garry McCracken and Peter Stott: *Fusion. The Energy of the Universe* (2nd ed. 2012)
- Hannes Alfvén: *Cosmic Plasma* (1981)
- EUROfusion: "JET's final fusion campaigns set a new energy record" (2024)

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